

Quant developer roadmap

Complete 12-month guide — what to do, what to read, and which chapters, in exact order

Key:

Book

Course

Project

Competition

Habit

Free

Paid

Phase 1 · Months 1–2

Build the foundation

1 Python for finance — get fluent fast

Work through Yves Hilpisch's book chapter by chapter. Type every code example into a Jupyter notebook — do not copy paste. After each chapter write a markdown cell explaining what you learned in your own words.

Read: Ch.1 (Why Python), Ch.2 (Setup), Ch.3 (Python basics), Ch.4 (NumPy), Ch.5 (pandas — most important), Ch.6 (OOP — skim if slow)

Book

Paid

2 Set up your GitHub and commit daily

Create your 'quant-learning' repo. Commit every notebook you make. The green contribution graph on your profile is checked by recruiters at Optiver and IMC — start building it from day one.

Project

Free

3 Read Flash Boys by Michael Lewis

Read this alongside the technical work — one chapter per night. Gives you the cultural and market context before the maths begins. Highly readable, zero equations.

Read: Whole book — 314 pages, ~5hrs total. Read cover to cover.

Book

Paid

4 Khan Academy — probability & statistics

Complete the full probability and statistics track. Do every exercise, not just the videos. Takes 3–4 weeks at 1hr/day. Probability is the core language of quant finance — everything else builds on it.

Complete: All modules in order — Probability, Counting, Random variables, Distributions, Hypothesis testing

Course

Free

Phase 1 reading schedule — months 1–2

Python for Finance — Hilpisch

Chapters 1–6

Ch.1 context, Ch.2 setup, Ch.3 Python syntax, Ch.4 NumPy, Ch.5 pandas (core), Ch.6 OOP

Flash Boys — Michael Lewis

Full book

Read cover to cover alongside technical work. One chapter per night.

Phase 2 · Months 3–4

Go deeper technically

5 The Concepts and Practice of Mathematical Finance — Joshi

Your first real quant textbook. Don't panic if it's hard — read it once cover to cover to build familiarity, then revisit chapters. Focus on understanding the intuition before the maths.

Read: Ch.1–4 (derivatives intro, arbitrage, risk-neutral pricing, Black-Scholes intuition). Skim Ch.5–8 first pass.

Book

Paid

6 Python for Finance — continued

Return to Hilpisch for the next set of chapters now that your Python fundamentals are solid. These chapters get directly into financial applications.

Read: Ch.7 (matplotlib visualisation), Ch.8 (financial time series — key chapter), Ch.9 (input/output)

Book

Paid

7 Quantopian lecture series — archived on GitHub

Work through every notebook in order. Covers factor models, portfolio construction, and risk in Python. The closest thing to a free structured quant curriculum. Search 'quantopian lectures github' to find it.

Complete: Notebooks 1–8 in sequence — statistics, factor models, portfolio theory, risk

Course

Free

8 Build your first backtesting project

Moving average crossover on ASX200 stocks using yfinance and pandas. Calculate returns, Sharpe ratio, max drawdown. Push to GitHub with a proper README. This is your first real CV talking point.

Project

Free

9 Options, Futures and Other Derivatives — Hull

The standard industry reference. Don't read the whole thing yet — work through the first 10 chapters which cover the foundations you need right now.

Read: Ch.1–3 (futures/forwards basics), Ch.4 (interest rates), Ch.5 (determination of forward prices), Ch.9–10 (options intro and properties)

Book

Paid

Phase 2 reading schedule — months 3–4

Python for Finance — Hilpisch

Chapters 7–9

Ch.7 visualisation, Ch.8 financial time series (most important in this block), Ch.9 data I/O

Concepts & Practice of Maths Finance — Joshi	Chapters 1–4
Full read of Ch.1–4, skim Ch.5–8 on first pass, return to them in Phase 3	
Options, Futures & Derivatives — Hull	Chapters 1–5, 9–10
Foundations of futures, forwards, interest rates, and options basics	
Quantopian lecture series	Notebooks 1–8
Work through in sequence on GitHub — free and practical	

10 IMC Prosperity Trading Challenge

Free algorithmic trading competition — build bots in Python competing in a simulated market. Run annually around March–April. Sign up the moment registrations open. Australian quant firms actively identify talent through this.

Competition

Free

11 Citadel / Citadel Securities Datathon

Data science and quant modelling competition. Teams analyse financial datasets and present models. Citadel Securities has a Sydney office and actively recruits from this competition.

Competition

Free

12 Read The Quants by Scott Patterson

The story of quant hedge funds before and during the 2008 crisis. Covers DE Shaw, Renaissance Technologies, Citadel. Read between competition prep sessions for motivation and cultural context.

Read: Full book — cover to cover. 352 pages. Read in evenings alongside competition prep.

Book

Paid

13 Joshi — deeper read of later chapters

Now that you have practical experience from building projects and entering competitions, return to Joshi's later chapters — they will make far more sense now.

Read: Ch.5 (the Black-Scholes formula), Ch.6 (risk and return), Ch.7 (exotic options — skim), Ch.8 (trees — important)

Book

Paid

14 Build a second more complex project

Upgrade your backtester with transaction costs, slippage, and position sizing — or build something new entirely: pairs trading strategy, a sentiment model using news headlines, or a Black-Scholes options pricer in Python.

Project

Free

Phase 3 reading schedule — months 5–6**The Quants — Scott Patterson****Full book**

Cover to cover. Read in evenings around competition prep.

Concepts & Practice of Maths Finance — Joshi**Chapters 5–8**

Return to Joshi now that you have practical context. Ch.5 Black-Scholes, Ch.8 trees are the key ones.

Hull — continued**Chapters 11–13**

Options pricing: binomial trees (Ch.13), Black-Scholes model (Ch.15 preview)

15 Heard on the Street — Timothy Crack

The classic quant interview prep book. Probability puzzles, brainteasers, and statistics questions used by Jane Street, Citadel, Optiver, and IMC. Do at least 5 problems per day on paper. Non-negotiable.

Work through: All probability chapters first, then statistics, then finance. Redo every question you got wrong after one week.

Book

Paid

16 A Practical Guide to Quantitative Finance Interviews — Xinfeng Zhou

Known as the Green Book. Covers the full range of quant interview question types. Work through alongside Heard on the Street — they complement each other perfectly.

Work through: Ch.1 (brain teasers), Ch.2 (probability — most important), Ch.3 (statistics), Ch.4 (stochastic calculus — do what you can)

Book

Paid

17 MIT OpenCourseWare — 18.S096 Mathematics for Finance

Free MIT lecture notes covering stochastic processes, Brownian motion, and the mathematical foundations of derivatives pricing. Work through the problem sets seriously.

Complete: Lecture notes 1–10. Do the problem sets — they are the same standard as quant interviews.

Course

Free

18 Python for Finance — advanced chapters

Return to Hilpisch for the final advanced section now that you have the mathematical foundation to understand what the code is doing.

Read: Ch.10 (performance Python), Ch.11 (mathematical tools), Ch.12 (stochastics — read alongside MIT notes)

Book

Paid

19 LinkedIn networking — 3 conversations per month

Search quant developer at Optiver Sydney, IMC Sydney, Macquarie, Susquehanna. Connect with people 2–3 years into their careers. Short, specific message about what you're studying and what you've built.

Habit

Free

Phase 4 reading schedule — months 7–9

Heard on the Street — Crack

Full book

Work through every problem. 5 per day minimum. Redo wrong answers after 1 week.

Quant Finance Interviews — Zhou (Green Book)

Ch.1–4

Brain teasers, probability (most important), statistics, intro stochastic calculus

Python for Finance — Hilpisch[Chapters 10–12](#)

Performance, mathematical tools, stochastics — read alongside MIT lecture notes

MIT OCW 18.S096[Lectures 1–10](#)

Free online. Work through problem sets — same standard as real quant interviews

20 Apply — Optiver, IMC, Susquehanna, Macquarie

The four primary targets for quant internships in Australia. Applications open mid-year for the following summer. Optiver and IMC run structured quant internship programs for penultimate-year students. Apply to all four.

Application

Free

21 Jane Street Electronic Trader Challenge or FTTT

Jane Street runs competitions and programs for students globally. Being selected for any Jane Street program is a strong CV signal recognised worldwide. Apply even if you feel uneasy.

Competition

Free

22 Paul Wilmott on Quantitative Finance — selective reading

The most comprehensive quant text in existence. Do not read cover to cover — use it as a reference. The chapters you read now will deepen your interview answers significantly.

Read selectively: Ch.3 (Black-Scholes derivation), Ch.7 (exotic options), Ch.14 (volatility surfaces), Ch.20 (numerical methods). Reference others as needed.

Book

Paid

23 Hull — Black-Scholes and Greeks deep dive

Return to Hull for the most important technical chapters. These underpin almost every derivatives pricing interview question at quant trading firms.

Read: Ch.15 (Black-Scholes-Merton model — read carefully), Ch.17 (the Greeks — essential), Ch.19 (volatility smiles — important)

Book

Paid

24 Daily market habit — maintain forever

AFR Markets section or The Daily Upside newsletter every morning (5 minutes). One Odd Lots or Acquired podcast per week. Follow quantian1 and nntaleb for quant thinking. Markets intuition compounds — start now.

Habit

Free

Phase 5 reading schedule — months 10–12

Paul Wilmott on Quantitative Finance

Selective chapters

Ch.3 Black-Scholes, Ch.7 exotics, Ch.14 vol surfaces, Ch.20 numerical methods — reference others

Hull — Options, Futures & Derivatives

Chapters 15, 17, 19

Black-Scholes model (Ch.15), the Greeks (Ch.17), volatility smiles (Ch.19) — read carefully

Heard on the Street — Crack

Revision

Re-do all problems you previously got wrong. Time yourself strictly.

Master reading summary

Every book, what to read, and when — at a glance

Python for Finance			Yves Hilpisch
Phase 1 — months 1–2	Ch.1–6	Core Python, NumPy, pandas. Type every example.	
Phase 2 — months 3–4	Ch.7–9	Visualisation, financial time series, data I/O.	
Phase 4 — months 7–9	Ch.10–12	Performance, mathematical tools, stochastics.	

The Concepts and Practice of Mathematical Finance			Mark Joshi
Phase 2 — months 3–4	Ch.1–4 (full read)	Derivatives intro, arbitrage, risk-neutral pricing.	
Phase 3 — months 5–6	Ch.5–8 (deeper read)	Black-Scholes, risk/return, trees. Makes more sense after projects.	

Options, Futures and Other Derivatives			John Hull
Phase 2 — months 3–4	Ch.1–5, 9–10	Futures, forwards, options basics.	
Phase 3 — months 5–6	Ch.11–13	Options pricing, binomial trees.	
Phase 5 — months 10–12	Ch.15, 17, 19	Black-Scholes, the Greeks, volatility smiles.	

Flash Boys			Michael Lewis
Phase 1 — months 1–2	Full book	Cover to cover. One chapter per night.	

The Quants			Scott Patterson
Phase 3 — months 5–6	Full book	Cover to cover. Read in evenings around competitions.	

Heard on the Street			Timothy Crack
Phase 4 — months 7–9	Full book — active practice	5 problems per day on paper. Redo wrong answers.	
Phase 5 — months 10–12	Revision	Re-do all previously wrong problems timed.	

A Practical Guide to Quant Finance Interviews			Xinfeng Zhou
Phase 4 — months 7–9	Ch.1–4	Brain teasers, probability, statistics, intro stochastics.	
Phase 5 — months 10–12	Ch.5–6 revision	Stochastic calculus and finance questions.	

Paul Wilmott on Quantitative Finance			Paul Wilmott
Phase 5 — months 10–12	Ch.3, 7, 14, 20 (selective)	Black-Scholes, exotics, vol surfaces, numerical methods. Reference only.	

